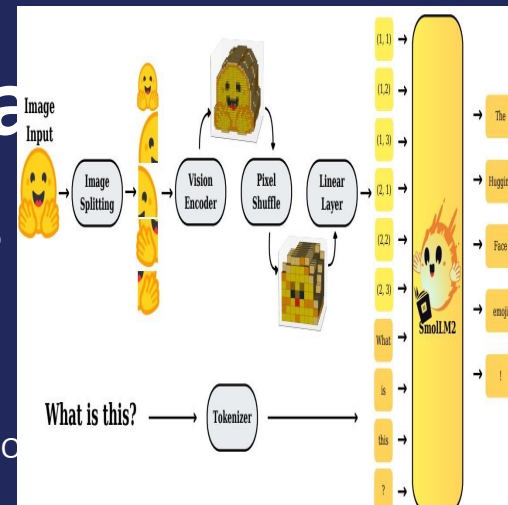


SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Sub-1GB inference. Competitive with models 300× larger. Fully open-source.



Large VLMs Are Powerful but Impractical for Edge Deployment

- State-of-the-art VLMs demand 10–80 GB GPU RAM
- Mobile and edge devices are limited to 4–16 GB total system memory
- Cloud dependency raises latency, cost, and privacy concerns
- Need: compact architectures that preserve capability at sub-1GB scale

27.7 GB

Molmo-A1B-7B RAM
Best efficient OS baseline

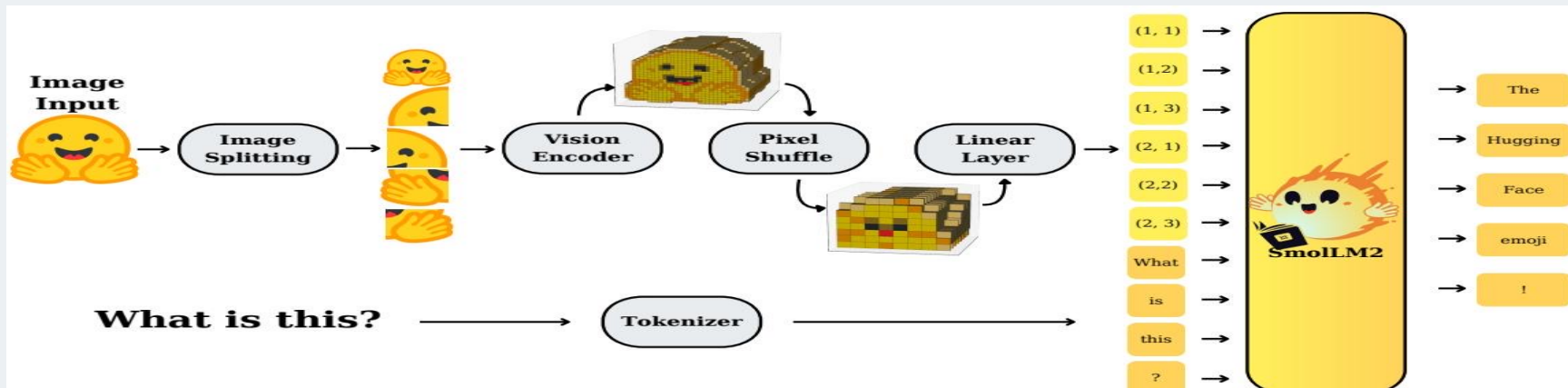
80 B

Idefics-80B Params
300× larger than SmolVLM-256M

0.8 GB

SmolVLM-256M RAM
Batch size 1, competitive accuracy

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



1

Image Splitting

Input image divided into sub-images for local encoding

2

Vision Encoder

SigLIP encoder extracts spatial feature maps

3

Pixel Shuffle

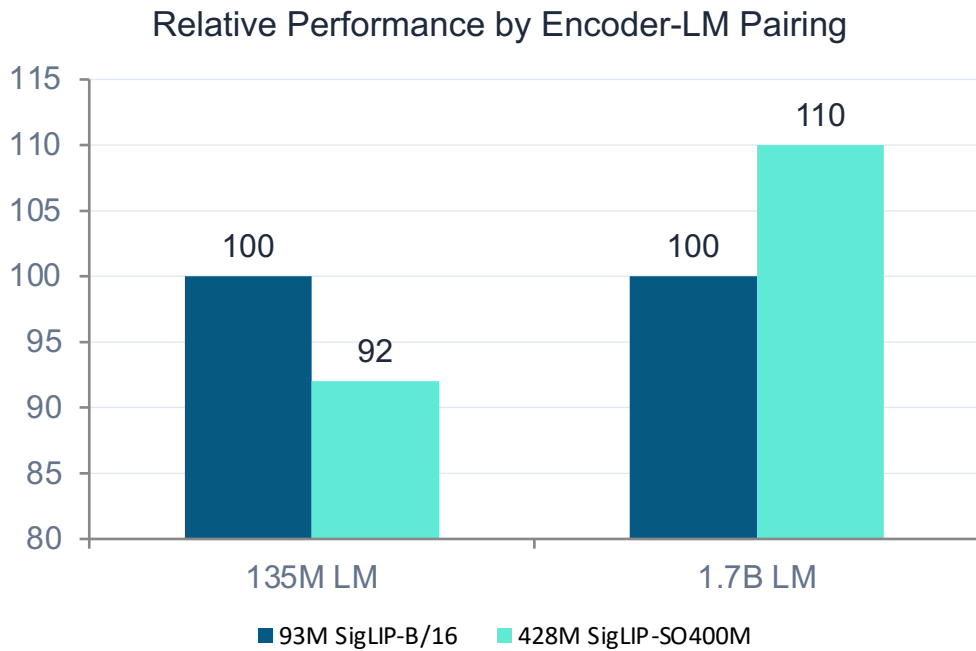
Space-to-depth rearrangement compresses tokens by r^2

4

SmolLM2 Fusion

Linear projection + text embeddings → language model

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



Finding 1

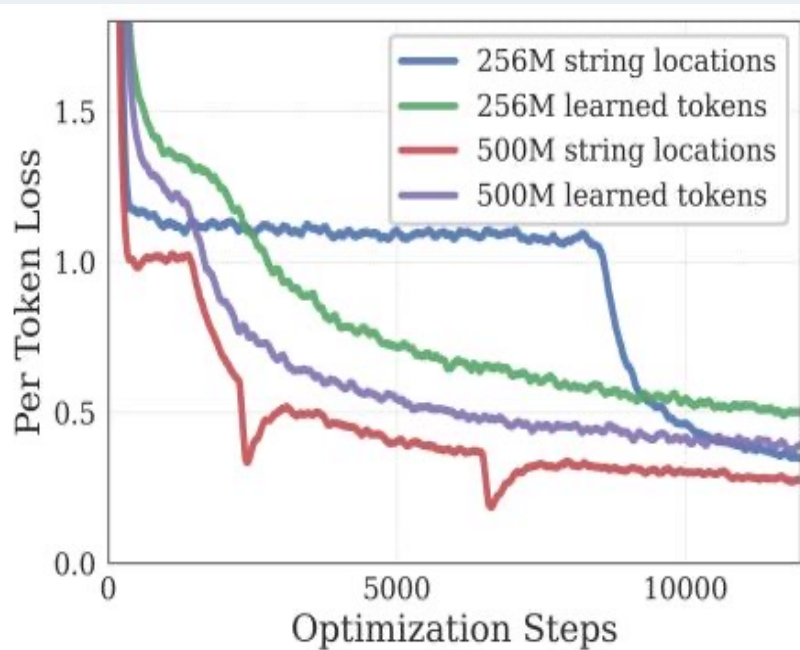
Balanced Encoder-LM Compute

A 93M SigLIP-B/16 paired with a 135M LM outperforms a 428M SigLIP-SO400M with the same LM.

Only at 1.7B LM scale does the larger encoder justify its ~10% parameter overhead.

Implication: allocate compute proportionally; oversized encoders hurt small VLMs.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



Mechanism

Spatial features are rearranged into depth channels, reducing token count by r^2 . At $r=4$, a 32×32 feature map becomes 8×8 — a 16× reduction.

Impact on Small VLMs

Larger models use $r=2$ (4× reduction) — sufficient with bigger LMs.

Small VLMs benefit from $r=4$ (16× reduction), easing attention overhead.

Enables longer context and higher effective resolution per token budget.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2

2K → 16K

Context length extension

10K → 273K

RoPE base adjustment

16K

SmolVLM-2.2B context

8K

256M / 500M context

How It Works

Increase RoPE base from 10,000 to 273,000 to extend positional encoding range.

Fine-tune on mixed long/short-context data to maintain stability across sequence lengths.

Performance improves significantly as context grows — more image patches and longer video frames fit within the token budget.

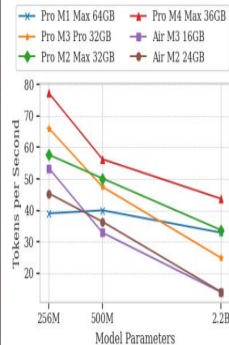
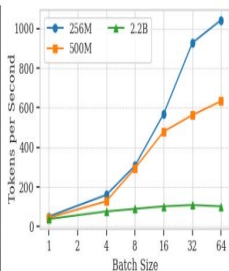
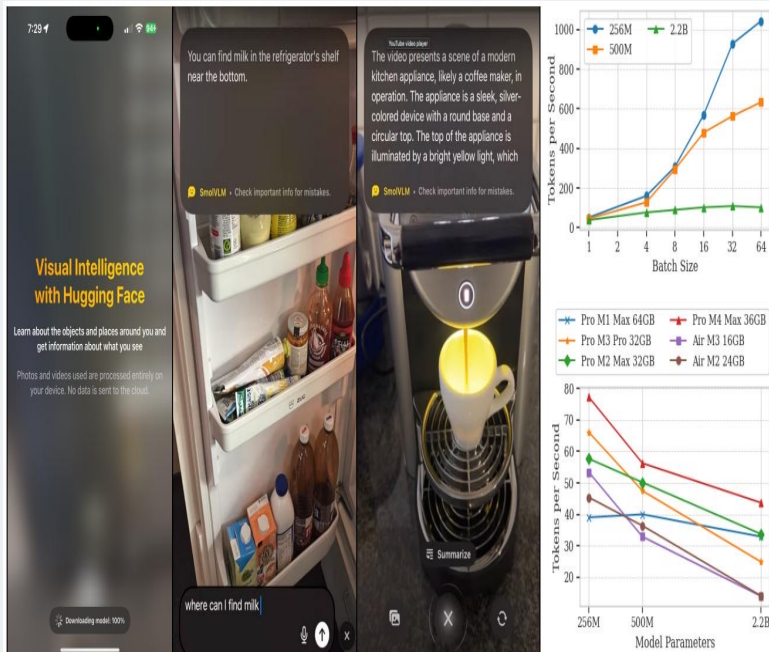
Trade-off: smaller variants (256M, 500M) use 8K context; only 2.2B scales to 16K.

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	256M	500M	2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average	All Benchmarks	44.0%	51.0%	59.8%	—

RAM Usage: 0.8 GB (256M) | 1.2 GB (500M) | 4.9 GB (2.2B) — vs. 27.7 GB for Molmo-A1B-7B

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



Throughput Benchmarks

~80 tok/s

SmoVLM-256M on Pro M1 Max 64GB

~1,000 tok/s

256M at batch size 64 (high-end devices)

~55 tok/s

256M on MacBook Air M3 16GB

Linear

Throughput scales roughly linearly with batch size

Key Takeaways: Design Principles for Efficient Multimodal Models

1

Balance Encoder-LM Compute

Pair small vision encoders with small LMs; scale encoder size only when the LM justifies it.

2

Extend Context Length

RoPE base scaling + mixed-context fine-tuning unlocks higher resolution and longer video.

3

Aggressive Token Compression

Small VLMs benefit from $r=4$ pixel shuffle (16× reduction), easing attention overhead.

4

Sub-1GB Inference Is Achievable

SmolVLM-256M uses 0.8 GB RAM and surpasses the 300× larger LLaVA-80B.

5

Open-Source for Reproducibility

All weights, datasets, code, and the HuggingSnap mobile app are fully open-sourced.